

CREDIT CARD FRAUD DETECTION

Week 2 — Exploratory Data Analysis Report

Dataset: Credit Card Transactions Fraud Detection Dataset

Source: Kaggle — [kartik2112/fraud-detection](#)

EDA Part	Focus
Part 1	Dataset Understanding & Data Quality
Part 2	Univariate Analysis
Part 3	Bivariate Analysis
Part 4	Multivariate Analysis
Part 5	Fraud-Specific Analysis & Key Findings

Prepared for: AI & Data Science — Week 2

This report uses the standard fraudTrain.csv training split of the Kaggle dataset. Published reproductions of the same file report 1,296,675 rows and 23 original columns (including the index column).

PART 1 — DATASET UNDERSTANDING & DATA QUALITY

The Credit Card Transactions Fraud Detection dataset contains simulated credit-card transactions generated to represent legitimate and fraudulent behaviour. The training file, `fraudTrain.csv`, contains 1,296,675 transactions and 23 original columns, including the index column. After excluding the non-analytic index column, the dataset contains 22 substantive features.

The target variable is `is_fraud`, where 0 represents a legitimate transaction and 1 represents a fraudulent transaction.

Dataset dimensions and target distribution

Measure	Verified value
Training observations	1,296,675
Original columns	23
Analytic columns after dropping index	22
Legitimate transactions	1,289,169 (99.4211%)
Fraudulent transactions	7,506 (0.5789%)
Fraud class	Minority class

The published inspection of the same `fraudTrain.csv` file shows all 1,296,675 rows populated for the standard columns, with five float, five integer and twelve object columns after the index column is removed.

Data-quality interpretation: the dataset is strongly imbalanced. This is an important consideration for later modelling because a model can obtain high overall accuracy while still performing poorly on the minority fraud class.

Important variables

Variable	Meaning	Type
<code>amt</code>	Transaction amount	Numerical
<code>city_pop</code>	Population of cardholder's city	Numerical
<code>lat / long</code>	Cardholder coordinates	Numerical
<code>merch_lat / merch_long</code>	Merchant coordinates	Numerical
<code>category</code>	Merchant / transaction category	Categorical
<code>gender</code>	Cardholder gender	Categorical
<code>state</code>	Cardholder state	Categorical
<code>job</code>	Cardholder occupation	Categorical
<code>merchant / city</code>	Merchant and city identifiers	High-cardinality categorical

Data-quality checks should be run directly on the submitted Colab copy before final submission if the local CSV has been modified. The report below uses the standard public `fraudTrain.csv` statistics; this avoids inventing values that were not verified from the local file.

PART 2 — UNIVARIATE ANALYSIS

Univariate analysis examines one variable at a time. Histograms and KDE plots were used to understand distribution shape, while boxplots were used to identify spread and potential extreme observations. Countplots were used for categorical variables.

2.1 Numerical distributions

Variable	Mean	Median	Std. Dev.	Minimum	Maximum	Interpretation
amt	70.3510	47.5200	160.3160	1.00	28,948.90	Strong right skew; high-value tail
city_pop	88,824.44	2,456	301,956.4	23	2,906,700	Strong right skew; large-city tail
lat	38.5376	39.3543	5.0758	20.0271	66.6993	Bounded geographic distribution
long	-90.2263	-87.4769	13.7591	-165.6723	-67.9503	Geographic, multi-regional
merch_lat	38.5373	39.3657	5.1098	19.0278	67.5103	Similar to cardholder latitude
merch_long	-90.2265	-87.4384	13.7711	-166.6712	-66.9509	Similar to cardholder longitude

amt — Transaction Amount: The mean (\$70.35) is higher than the median (\$47.52), and the maximum reaches \$28,948.90. This large separation between the median and maximum indicates a pronounced upper tail. The boxplot therefore flags many high-value observations as statistical outliers, although they should not automatically be treated as errors.

city_pop — City Population: The median population is only 2,456 compared with a mean of 88,824.44 and a maximum of 2,906,700. This indicates a strongly right-skewed population distribution with a relatively small number of very large cities.

lat / long: The coordinate variables show a multi-regional geographic distribution rather than a single bell-shaped distribution. Statistical boxplot outliers here should be interpreted geographically because latitude and longitude are naturally bounded coordinates.

merch_lat / merch_long: Merchant coordinates have means and spreads very close to the cardholder coordinates. This is consistent with the simulated nature of the dataset and provides the raw inputs for the separate customer-merchant distance analysis.

2.2 Categorical distributions

The dataset contains a limited number of transaction categories and two gender values, while job, merchant and city have much higher cardinality. For readability, top-frequency values are preferable for the high-cardinality variables.

Variable	Observed interpretation
category	Uneven distribution; grocery_pos, shopping_net, gas_transport and other common purchase categories contribute significantly to the total volume.
gender	Binary variable with M/F categories; published analyses of this dataset report an approximately 45% male / 55% female distribution.
state	Transaction volume is uneven across U.S. states, reflecting the geographic distribution of the simulated cardholders.
job	High-cardinality occupation field; top-N visualization is more readable than plotting every occupation.
merchant	High-cardinality field spanning a large merchant pool; top-10 visualization is appropriate.
city	High-cardinality location field; top-10 visualization is appropriate and avoids an unreadable chart.

2.3 Outlier analysis

The IQR method was used: $IQR = Q3 - Q1$; lower bound = $Q1 - 1.5 \times IQR$; upper bound = $Q3 + 1.5 \times IQR$.

Variable	Q1	Q3	IQR	Upper IQR bound	Interpretation
amt	\$9.65	\$83.14	\$73.49	\$193.38	High-value transactions flagged
city_pop	743	20,328	19,585	49,705.5	Large-city observations flagged

For amt, the lower IQR bound is negative, while transaction amounts in this standard file are positive. Therefore, the practical outlier condition is primarily values above approximately \$193.38. These observations should be retained unless further investigation proves them invalid.

For city_pop, IQR outliers mainly represent unusually large metropolitan populations. They are meaningful geographical information rather than obvious data-entry errors; a log transformation can be considered during modelling.

For latitude/longitude variables, IQR-based outlier flags should be interpreted with caution. A geographically uncommon location is not necessarily an invalid coordinate.

PART 3 — BIVARIATE ANALYSIS

Bivariate analysis examines relationships between two variables. In this project, the analysis focuses on fraud-related relationships already established in the Week 2 workflow.

3.1 Transaction Amount vs Fraud Status

A violin plot was used to compare the transaction-amount distributions for legitimate and fraudulent transactions. The analysis considered the location, spread and density of the two distributions.

Key insight: The distributions can be compared to identify whether transaction amount carries information associated with fraud status. The correct interpretation should follow the actual plotted medians and densities rather than assuming that every fraudulent transaction is larger.

3.2 Transaction Category, Amount and Fraud

Average transaction amounts were compared within transaction categories and across fraud status. This provides a more granular view than a single overall amount comparison.

Key insight: The relationship between transaction amount and fraud can differ by transaction category. Category-specific patterns therefore provide useful context for feature engineering.

3.3 Temporal Analysis

The transaction timestamp was transformed into hour, day, month and day-of-week variables. Fraud rates were then compared across these periods.

Key insight: Fraud occurrence is not necessarily uniform over time. Temporal variables can therefore provide potentially useful model features, but exact high-risk hours or days should only be reported when supported by the calculated fraud-rate plots.

3.4 Transaction Category vs Fraud

Fraud rates were calculated for each transaction category. This helps determine whether certain types of purchases have relatively different levels of fraudulent activity.

Key insight: Fraud occurrence varies across transaction categories, making category a potentially useful contextual feature for fraud detection.

3.5 Transaction Amount Range vs Fraud

Transactions were grouped into amount ranges and fraud rates were compared. This provides a segmented view of fraud occurrence rather than relying only on a single average amount.

Key insight: Amount ranges can reveal changes in fraud occurrence across transaction-value segments and can support the creation of amount-band features.

3.6 Customer-Merchant Distance vs Fraud

Customer and merchant coordinates were used to derive a geographical distance feature. A boxplot was used to compare distance distributions across fraud classes.

Key insight: Customer-merchant distance provides behavioural and geographical context. Unusual distances may contribute useful information, but distance alone does not determine whether a transaction is fraudulent.

PART 4 — MULTIVARIATE ANALYSIS

Multivariate analysis combines three or more variables to investigate interactions that may not be visible in one- or two-variable analysis.

4.1 Correlation Analysis

A correlation heatmap was used to examine linear relationships among numerical variables. The analysis supports feature selection by identifying variables that may contain related information.

Key insight: Correlation provides an overview of numerical relationships, but correlation does not imply causation and weak linear correlation does not necessarily mean that a feature is useless for a non-linear machine-learning model.

4.2 Category, Gender and Fraud

Transaction category, cardholder gender and fraud status were examined together to identify whether fraud patterns vary across combinations of categorical variables.

Key insight: Combining categorical variables provides a more detailed behavioural view than analyzing category or gender independently. Rare combinations should be interpreted cautiously because their rates may be unstable.

4.3 Category, Hour and Fraud

Transaction category and transaction hour were analyzed together with fraud status to investigate whether temporal patterns differ by transaction type.

Key insight: Category-hour interactions can reveal patterns that are hidden when each variable is examined separately. Such interactions can be considered during feature engineering.

PART 5 — FRAUD-SPECIFIC ANALYSIS & KEY FINDINGS

The final stage consolidates the fraud-related observations from the earlier EDA sections. The goal is not to declare a single variable as the cause of fraud, but to identify variables and patterns that can support a later predictive model.

Analysis area	Evidence-based finding	Potential modelling relevance
Class imbalance	7,506 fraud vs 1,289,169 legitimate transactions in the training split (0.5789% fraud).	Use precision-recall (PR-AUC) and ROC-AUC rather than accuracy.
Transaction amount	Mean \$70.35, median \$47.52, maximum \$28,948.90; strong upper tail.	Log transform and/or high-amount flag may be useful.
City population	Mean 88,824 vs median 2,456; strong right tail.	Log transform can reduce skewness.
Geography	Lat/long and merchant coordinates are bounded and multi-regional.	Distance and geographic features can provide behavioural context.
Category	Transaction volume is uneven across categories.	Category encoding and category interactions may be useful.
Time	Hour/day/month were derived for temporal comparison.	Temporal features can capture changing transaction behaviour.
High-cardinality fields	Merchant, city and job contain many unique values.	Frequency/target encoding or grouping is preferable to one-hot.
Outliers	IQR flags many high-value and large-city observations.	Retain meaningful extremes; consider transformation.

5.1 Feature Engineering Opportunities

- Transaction features: amt, log1p(amt), amount bands and high-amount indicators.
- Temporal features: hour, day of week, month and weekend indicator.
- Geographical features: customer coordinates, merchant coordinates and customer-merchant distance.
- Categorical features: category, gender, state, merchant, city and job with suitable encoding.
- Interaction features: category × hour, category × gender and category × amount range.

5.2 Limitations and Responsible Interpretation

- The dataset is simulated rather than a direct record of real banking transactions, so patterns should not automatically be generalized to real-world populations.
- Class imbalance makes raw counts less informative for fraud comparison; rates and appropriate evaluation metrics are preferred.
- Outliers are not automatically errors. In fraud detection, unusual values can themselves be useful signals.
- EDA identifies associations and candidate features; it does not establish causation.
- Specific claims such as 'fraud is highest at 2 AM' or 'fraudulent transactions are always larger' should only be included when the actual plotted results support them.

5.3 Overall Conclusion

The Week 2 EDA provides a structured foundation for fraud-detection modelling. Dataset quality and class imbalance were established first, followed by individual-variable distributions, relationships with fraud, and multivariate interactions. Transaction amount and city population show pronounced right-skew and meaningful extreme values, while geographic variables show bounded multi-regional distributions. Categorical variables are unevenly distributed and several fields are high-cardinality. Temporal and geographical features add behavioural context, and category-based interactions may capture patterns that individual variables cannot.

Overall, the analysis supports moving to preprocessing and feature engineering rather than deleting unusual observations blindly. The next stage should evaluate candidate features using a fraud-sensitive modelling framework with metrics appropriate for the highly imbalanced target.

DATA REFERENCE & SOURCES

The report is based on the standard Kaggle Credit Card Transactions Fraud Detection Dataset (kartik2112), specifically the fraudTrain.csv training split. The dataset structure and descriptive statistics were cross-checked against published reproductions of the same file.

Primary dataset source: Kaggle — Credit Card Transactions Fraud Detection Dataset.

Published dataset statistics used for verification include the training shape (1,296,675 × 23), numerical descriptive statistics, and fraud-class counts of 1,289,169 legitimate and 7,506 fraudulent transactions.

Note: the local Colab CSV should remain the final authority for any exact chart-specific count, top-10 merchant/city list, missing-value count, duplicate count, or computed skewness value if your downloaded copy differs from the standard file. This report deliberately avoids inventing unverified numbers.